Task Sufficiency Does Not Certify Quotient Faithfulness

A Theory of What Standard Criteria Cannot Verify

Peter Farkas
Åbo Akademi University, DAZE Project

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Abstract

A learned representation that achieves perfect task accuracy need not encode the equivalence structure of the data-generating process. Standard criteria such as samplewise predictive fit, sufficient-statistic characterisations, and information-bottleneck objectives evaluate a representation only relative to a prediction target; they do not in themselves determine whether the representation respects a declared semantic quotient. This paper formalises that gap.

Three logically distinct limitations are established. First, any finite labelled sample admits predictive solutions whose latent organisation does not reflect the underlying equivalence structure: a one-dimensional traversal of the sample cloud always achieves perfect training accuracy, providing no evidence that intrinsic quotient structure has been recovered. This is a statement about the absence of distinguishing signal in the criterion, not about the likelihood of such solutions under any particular optimiser. Second, when the task map is non-injective on the declared quotient, maximal task-sufficient compression collapses distinctions that quotient-faithful encoding would need to preserve. This is a claim about the structure of the compression target—the global minimum of the compression objective—not about the trajectory followed by any particular optimiser. Third, once a quotient space $\mathcal{Q}$ is declared, any topology-faithful encoding must satisfy geometric embedding constraints imposed by $\mathcal{Q}$ itself; these constraints are external to predictive objectives.

The paper does not claim that standard training can never recover quotient-faithful structure. Its narrower conclusion is that task performance and task sufficiency alone do not certify such recovery. Any claim that a learned representation faithfully encodes an underlying quotient therefore requires additional structure beyond predictive success—an explicit admissibility declaration, an independently justified structural prior, or both.

About this series

A learned representation that achieves perfect task accuracy has not necessarily encoded the equivalence structure of the data-generating process. The series Quotient Faithfulness in Representation Learning develops a unified account of when and why this gap arises and what is required to close it. A representation is quotient-faithful *if it maps semantically equivalent inputs to the same code and faithfully encodes the space of semantically distinct states as a topological structure in the latent space. Three structural traps prevent standard training from achieving this: the traversal trap, the compression trap, and the dimension trap.*

Paper 1 *[Farkas, 2026a] (this paper): Establishes the three traps and proves that standard training criteria cannot certify quotient faithfulness.*

Paper 2 *[Farkas, 2026b]: Identifies three recovery routes and four admissibility sources. Establishes the three-zone compression trap structure and its Kolmogorov complexity foundations.*

Paper 3 *[Farkas, 2026c]: Derives geometric lower and constructive upper bounds on the latent space dimension required for topology-faithful encoding.*

Paper 4 *[Farkas, 2026d]: Maps the framework against six major traditions in representation learning theory and examines fourteen bodies of experimental evidence.*

Paper 5 *[Farkas, 2026e]: Shows that partial structural knowledge — generator specifications — constrains the admissible quotient class and imposes certifiable architecture floors without naming the equivalence relation explicitly. Introduces the Generator-Constrained Discovery (GCD) scheme.*

Paper 6 *[Farkas, 2026f]: Empirical validation of the GCD framework on benchmark dynamical systems: phase cylinder, Hopf fibration, and double pendulum. Benchmarks against Hamiltonian Neural Networks [Greydanus et al., 2019].*

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Notation

Symbol	Meaning
$\mathcal{X}$	Observation space
$\sim$	Admissible nuisance relation on $\mathcal{X}$
$\mathcal{Q} = \mathcal{X}/\sim$	Lawful quotient: space of semantic states
$q : \mathcal{X} \to \mathcal{Q}$	Canonical quotient projection
Y	Target label space
$f : \mathcal{X} \to Y$	Prediction target ($f = t \circ q$)
$t : \mathcal{Q} \to Y$	Task map (factor of f through $\mathcal{Q}$)
Z	Exposed latent space
$E : \mathcal{X} \to Z$	Encoder (representation map)
$\tilde{E} : \mathcal{Q} \to Z$	Induced quotient representation
E_{label}	Label representation: $E_{\text{label}}(x) = f(x) = t(q(x))$
$E_1 \preceq E_2$	E_1 at least as compressed as E_2 (compression preorder)
$\dim(\mathcal{Q})$	Covering dimension of $\mathcal{Q}$
$\text{emb}(\mathcal{Q})$	Minimum Euclidean embedding dimension of $\mathcal{Q}$
S^1	Circle (canonical non-contractible quotient example)
$\Sigma(\mathcal{Q}, Z)$	Discriminant: boundary of C^1 embeddings $\mathcal{Q} \hookrightarrow Z$

1 Introduction

Position in the series. A learned representation that achieves perfect task accuracy has not necessarily encoded the equivalence structure of the data-generating process. This paper is the first in the series *Quotient Faithfulness in Representation Learning* and establishes the theoretical foundation that the subsequent papers build on: three structural traps that prevent standard training criteria from certifying—and often actively undermine—quotient-faithful representations. Papers 2 [Farkas, 2026b] and 3 [Farkas, 2026c] address recovery and architecture respectively; Paper 4 [Farkas, 2026d] maps the framework against the broader theoretical landscape.

This paper. This paper proves that standard training criteria cannot certify quotient faithfulness. A representation can achieve perfect predictive accuracy, satisfy the minimal sufficient statistic criterion, and optimise any information-bottleneck objective—and still fail to capture the equivalence structure of the data-generating process. Three independent theorems establish this. No refinement of these criteria alone resolves the problem, because they remain blind to nuisance structure unless additional admissibility information is supplied.

The three results form a sequence that tightens the impossibility. Corollary 3.3 (Traversal Ambiguity) opens the door: for any finite sample, there exists a one-dimensional interpolation of the data that achieves perfect training accuracy. The samplewise objective provides no signal that distinguishes a quotient-faithful representation from this traversal. This is an existence and indistinguishability result: the criterion cannot rule out such

representations, not a claim that optimisers reliably find them. Theorem 4.4 (Task-Map Collapse) is the central result: the minimal sufficient statistic—the global optimum of the compression objective—induces the task map t on the semantic quotient $\mathcal{Q}$, and therefore collapses $\mathcal{Q}$ whenever t is non-injective. This is not a local minimum or an artefact of finite data: the global optimum of the compression objective is a structurally wrong point, and no task-based criterion can detect this. This is a claim about the loss landscape, not the training trajectory. Theorems 7.2 and 7.4, and Corollary 7.5 (Embedding Obstructions), complete the obstruction: once the quotient $\mathcal{Q}$ is fixed, the latent space must embed $\mathcal{Q}$ topologically and metrically. These constraints follow from the geometry of $\mathcal{Q}$ itself and apply to any induced map $\tilde{E} : \mathcal{Q} \to Z$ once the latent space Z is fixed.

Throughout this paper, a single running example grounds the formal results. Consider a cyclic dynamical process whose observable $x(t) \in \mathbb{R}^m$ is driven by a phase $\theta(t) \in S^1 = \mathbb{R}/2\pi\mathcal{Z}$, with an irrelevant amplitude nuisance making the observation space a cylinder $C = S^1 \times [0, 1]$. The declared quotient is $q : C \to S^1$. The prediction target is the sine projection $t(\theta) = \sin(\theta)$, which is non-injective: phases θ and $\pi - \theta$ produce the same output. Phase processes arise throughout science and engineering—oscillators, circadian rhythms, rotational systems, heading in navigation—and the circle $\mathcal{Q} = S^1$ is the simplest compact topological space that is not contractible, making it the canonical test case for all three failure modes.

This paper introduces two formal tools: the *compression preorder* ($\preceq$) on sufficient representations, which makes the partial order structure of sufficiency precise; and the *quotient faithfulness criterion*, which requires the induced map $\tilde{E} : \mathcal{Q} \to Z$ to be a topological or metric embedding. These tools are the vocabulary in which the three impossibility results are stated. The paper addresses certification, not discovery: it characterises necessary conditions for lawful abstraction and shows that standard evaluation criteria cannot substitute for an explicit declaration of admissible equivalences.

Supervised losses, contrastive objectives, and information-bottleneck variants do not, by themselves, certify quotient structure—the theorems explain why. The closest related bodies of work each illuminate part of the problem without resolving it. The information bottleneck [Tishby and Zaslavsky, 2015] proposes the minimal sufficient statistic as the ideal representation; the results below show this is precisely the representation that destroys quotient structure when the task map is non-injective, which is the common case. Locatello et al. [2019] prove that disentanglement is unidentifiable without inductive biases beyond the training objective; Schölkopf et al. [2021] argue that causal invariances supply those biases. Both results point to the same gap this paper formalises: structural constraints require a declaration beyond predictive fit. Bengio et al. [2013] articulate the goal of representation learning as recovering the underlying factors of variation; the present paper asks whether standard criteria certify that it has been achieved.

Recent foundational work by Kvalheim and Sontag [2024] establishes critical topological constraints on representation learning, specifically proving that a perfect autoencoder with a bottleneck of $\mathbb{R}^k$ exists if and only if the data manifold is homeomorphic to a subset of $\mathbb{R}^k$. While their work provides strictly stronger mathematical guarantees regarding topological embeddings, the present paper shifts the focus from autoencoder realizability to a broader evaluation critique. Our contribution is not stronger theorems, but rather a unified diagnostic framework that connects classical topology and statistics to the representation learning evaluation problem—and two novel formalisations (the compression preorder and the quotient faithfulness criterion) that give precise language for the gap that stronger

theorems alone cannot supply.

The present paper sharpens the Locatello et al. [2019] result in a specific direction. Locatello et al. show that the correct factorisation cannot be identified from data alone; Theorem 4.4 shows that even when the admissibility structure is explicitly declared, the global optimum of the compression objective sits at a structurally wrong point with respect to it. The impossibility here is not about identification under ambiguity—it is about what the global optimum of compression does to a known quotient. The two results are logically independent: one concerns the epistemic problem of selecting $\mathcal{Q}$ from data; the other concerns where the global optimum of the compression objective sits once $\mathcal{Q}$ is declared.

Equivariant architectures [Cohen and Welling, 2016, Bronstein et al., 2021] resolve the problem in one regime: when the admissible nuisance relation is a known group action. This paper addresses what happens outside that regime. Different architecture families implicitly encode admissibility priors—convolutional networks enforce translation consistency, Transformers assume token exchangeability, graph networks impose relational structure, contrastive methods define equivalence through augmentation—but none of these, as currently evaluated, certifies that the encoded prior matches the true nuisance structure. The present results characterise precisely what is left uncertified in all of these cases.

> *A compressed sufficient representation should not be interpreted as a lawful abstraction unless the source of admissibility is explicitly stated.*

These results should not be read as claiming that neural networks cannot recover lawful abstractions in practice. They can and often do. The point is narrower: when that happens, it cannot be credited to samplewise fit, sufficiency, or compression alone. Some additional structure must be doing the work—for example architectural priors, augmentation-induced equivalences, stochastic optimization bias, neighbourhood objectives, or other implicit admissibility signals. The present paper identifies the formal gap; the constructive question of how practical systems sometimes bridge it is deferred.

The paper, and the series it begins, is at its root a theory of the gap between *task-relative adequacy* and *lawful abstraction*: the gap between representations that achieve predictive success and representations that faithfully encode the equivalence structure of the data-generating process. Traversal ambiguity, compression collapse, and geometric impossibility are three distinct ways in which that gap manifests; the admissibility declaration is the structural variable needed to close it.

Proofs of classical results are in Appendix B; theorem attribution in Appendix C; AI disclosure in Appendix D. The first author takes full scientific responsibility for all claims.

2 Formal Setting: Lawful Quotients and Quotient-Respecting Representations

The goal of representation learning, as articulated by Bengio et al. [2013], is to recover the underlying factors of variation from data. The question this paper asks is whether standard training criteria certify that it has been achieved. The information bottleneck [Tishby and Zaslavsky, 2015] proposes the minimal sufficient statistic as the ideal representation; the results below show this is precisely the representation that destroys factor structure when the task map is non-injective. Locatello et al. [2019] prove that disentanglement

is unidentifiable without inductive biases beyond the training objective; Schölkopf et al. [2021] argue that causal invariances supply those biases. Both results converge on the same requirement: structural constraints on representations require a declaration of which equivalences are semantically valid, and standard objectives cannot substitute for that declaration.

Equivariant architectures [Cohen and Welling, 2016, Bronstein et al., 2021] resolve the problem in one regime: when the admissible equivalences form a known group action. Different architecture families handle adjacent regimes implicitly—convolutional networks enforce translation consistency, Transformers assume token exchangeability, graph networks impose relational structure, contrastive methods define equivalence through augmentation—but in each case the encoded prior is present by construction rather than certified by evaluation. This section introduces the formal language that makes the gap precise: what it means to declare an admissible equivalence, what structure the declaration induces, and what a representation must do to honour it.

The setting is deliberately minimal. Every concept used in Sections 2 through 7 is defined here. The response to the limits established in Sections 3–7 is not a better loss function or a deeper architecture—it is a declaration of which equivalences are semantically admissible. This section formalises that declaration.

2.1 Observation Space and Admissible Nuisance

Let X be the observation space. In standard representation learning X is the space of raw inputs: images, sequences, sensor readings, or any other domain from which training data are drawn.

Definition 2.1 (Admissible nuisance relation)**.** An *admissible nuisance relation* is an equivalence relation $\sim$ on X that is declared or otherwise justified as identifying observations that differ only in semantically irrelevant ways. We write $x \sim x'$ when x and x' are equivalent under $\sim$, and call the equivalence classes *nuisance fibers*.

The relation $\sim$ is not discovered from data by any procedure described in this paper; it is given externally. In practice it may come from known symmetries of the domain, from physical invariances, from multiview supervision, from causal interventions, or from domain knowledge. The paper's scope is confined to consequences of having declared $\sim$, not to the problem of finding it. A forthcoming follow-up paper (Farkas, in preparation) addresses realistic sources of admissibility.

2.2 The Lawful Quotient

Given an admissible nuisance relation $\sim$ on X, the quotient space $\mathcal{Q} = X/\sim$ is the set of equivalence classes. The canonical projection

$$q : X \to \mathcal{Q}, \quad q(x) = [x]$$

maps each observation to its equivalence class.

Definition 2.2 (Lawful quotient)**.** The pair $(\mathcal{Q}, q)$ where $\mathcal{Q} = X/\sim$ and $q : X \to \mathcal{Q}$ is the canonical projection is called the *lawful quotient* induced by $\sim$. The space $\mathcal{Q}$ represents the semantic structure that remains after removing nuisance variation.

The quotient $\mathcal{Q}$ is the space of semantic states: the distinct configurations that matter for the task after all nuisance variation has been factored out. A representation that

faithfully encodes $\mathcal{Q}$ is one that assigns distinct codes to semantically distinct states and the same code to semantically equivalent ones, while preserving the topological and geometric relationships among semantic states.

2.3 The Prediction Target

Assumption 2.3 (Quotient-compatible target)**.** There exists a measurable map $t : \mathcal{Q} \to Y$ such that

$$f = t \circ q.$$

That is, f depends on x only through its equivalence class $q(x)$.

This is a minimal coherence condition between the nuisance declaration and the task. Note that t need not be injective. This non-injectivity of t is precisely what creates the gap between task sufficiency and quotient faithfulness that Theorem 4.4 exploits.

2.4 Representation Maps and the Exposed Latent Space

Definition 2.4 (Representation map)**.** A *representation map* (or encoder) is a measurable map $E : X \to Z$ where Z is called the *exposed representation space* (or latent space). In most neural architectures $Z = \mathbb{R}^m$ for some finite m. All structural results in this paper concern the map $E : X \to Z$ as a whole and the induced map $\tilde{E} : \mathcal{Q} \to Z$ it produces. They make no claims about the internal layer structure of any network that implements E.

2.5 Quotient-Respecting Representations

Definition 2.5 (Quotient-respecting representation)**.** A representation $E : X \to Z$ *respects the quotient* $q : X \to \mathcal{Q}$ if there exists a measurable map $\tilde{E} : \mathcal{Q} \to Z$ such that

$$E = \tilde{E} \circ q.$$

The map $\tilde{E}$ is called the *induced quotient representation.*

The factorisation $E = \tilde{E} \circ q$ means that $E(x)$ depends on x only through its equivalence class $q(x)$.

Remark 2.6 (Representations that do not respect the quotient). If E does not factor through q, there exist $x, x' \in X$ with $q(x) = q(x')$ but $E(x) \neq E(x')$. Such a representation is sensitive to nuisance variation by construction and is therefore outside the class of lawful abstractions considered in this paper. This is a hard boundary, not a matter of degree.

2.6 Reduction to the Induced Map

Observation 2.7. Let $E : X \to Z$ be a quotient-respecting representation with induced map $\tilde{E} : \mathcal{Q} \to Z$. Then:

(a) Whether E preserves the topology of $\mathcal{Q}$ depends entirely on whether $\tilde{E}$ is a topological embedding.
(b) Whether E preserves the metric geometry of $\mathcal{Q}$ depends entirely on whether $\tilde{E}$ is bi-Lipschitz.
(c) The properties studied here concern $\tilde{E} : \mathcal{Q} \to Z$ once a representation has been formed. Questions of realizability, approximation, and learnability are addressed in the forthcoming follow-up paper (Farkas, in preparation).

The embedding obstruction theorems are statements about $\tilde{E} : \mathcal{Q} \to Z$, and they hold regardless of how E is computed. The expressiveness of the architecture does not help if Z cannot topologically accommodate $\mathcal{Q}$.

2.7 What This Framework Does and Does Not Address

This paper does not propose a method for discovering the admissible nuisance relation $\sim$ from data. The framework assumes $\sim$ is given.

The framework does not assume that $\mathcal{Q}$ is a manifold. The quotient space $\mathcal{Q}$ is an abstract topological space; the results use compactness and covering dimension, which apply to general metric spaces.

The framework treats $\mathcal{Q}$ as a single quotient space defined on all of X by a single equivalence relation $\sim$. Many naturally occurring problems do not have this structure: their quotient map varies across regions of the domain, with distinct per-region quotients glued on overlaps or separated by discrete state switches. Examples include dynamical systems whose phase space stratifies by energy with bifurcating fibre topology (double pendulum, Paper VI §7), systems with discrete control regimes whose operative dynamics differ per regime (Color Fantasy ship hotel, Paper VI §8), hybrid control systems, and thermodynamic systems across phase transitions. The generalisation to regionally varying quotient structure — a *quotient atlas* — is developed in Paper 7 (forthcoming). The non-certification results of this paper extend to quotient atlases with additional failure modes: task success can be achieved with different unfaithful encodings per region, undetectable from within-region evaluation alone. Within this paper we restrict attention to the single-quotient case.

The framework assumes that the latent space Z is in principle capable of representing $\mathcal{Q}$. Section 7 characterises the minimum requirements on Z for faithful encoding, while Sections 3 and 4 show how standard objectives can fail to use an adequate Z faithfully.

A governing distinction runs throughout the paper. The structural results of Sections 3–7 are claims about $\tilde{E} : \mathcal{Q} \to Z$ given a fixed latent space Z. These results are independent of architecture. The realizability results—lower bounds on width, upper bound questions, the architecture complexity programme—are addressed in the forthcoming follow-up paper (Farkas, in preparation).

A related clarification concerns the difference between the exposed representation space and internal architecture. The framework treats $E : X \to Z$ as a black box: it cares only about the map from inputs to exposed representations, not about the layers, widths, or internal bottlenecks of the network that computes it. The dimension lower bound of Theorem 7.4 is a constraint on $\dim(Z)$, not on any internal layer width. A reader should not infer that a network with any layer of width $k < \dim(\mathcal{Q})$ cannot produce a topology-faithful representation in $Z = \mathbb{R}^m$ with $m \geq \dim(\mathcal{Q})$: the internal architecture may expand, project, or transform representations in arbitrary ways before producing the exposed output. The question of which internal architectures can realise a faithful $\tilde{E}$ is a separate realizability question outside the scope of this paper.

3 Finite-Sample Ambiguity and the Limits of Predictive Fit

Before we can ask whether a representation faithfully encodes a quotient structure, we should ask a simpler question: could finite training data have forced any representation to encode it? The answer is no.

3.1 The Finite-Sample Traversal Theorem

Theorem 3.1 (Finite-sample traversal; classical: Whyburn 1942, Rourke and Sanderson 1972)**.** *Let $D = \{x_1, \ldots, x_n\} \subset \mathbb{R}^m$ be a finite set of pairwise distinct points. Then there exists a continuous injective map*

$$\gamma : [0, 1] \to \mathbb{R}^m$$

and parameters $0 < t_1 < \cdots < t_n < 1$ such that $\gamma(t_i) = x_i$ for all $i = 1, \ldots, n$.

This is a classical result in piecewise-linear topology. Proof via a generic projection direction and piecewise-linear interpolation is in Appendix B.

Remark 3.2 (Genericity of the construction)*.* The direction v used in the proof is generic: it suffices to avoid a measure-zero set of hyperplanes. No special structure of D is required.

3.2 Samplewise Fit Does Not Identify Intrinsic Structure

Corollary 3.3 (Traversal Ambiguity)**.** *Let $D = \{x_1, \ldots, x_n\} \subset X \subset \mathbb{R}^m$ be a finite training set drawn from some lawful quotient structure $q : X \to \mathcal{Q}$, and let y_i be arbitrary target values. Then there exists a one-dimensional coordinate assignment $\tau : D \to [0, 1]$ and a readout $h : [0, 1] \to Y$ such that $h(\tau(x_i)) = y_i$ for all i.*

Proof. Immediate from Theorem 3.1. See Appendix B. □

The corollary means that the samplewise objective has no mechanism to distinguish a representation induced by the true lawful quotient q from a one-dimensional traversal of the sample cloud. Theorem 3.1 is an existential ambiguity result, not a realizability theorem: it does not claim that any specific network can learn the traversal, only that the samplewise objective provides no basis for ruling it out.

3.3 Illustration: Traversal Ambiguity on the Cylinder

The observation space is $C = S^1 \times [0, 1]$, the quotient is $q : C \to S^1$, and the learner receives a finite sample D with targets $y_i = \sin(\theta(x_i))$. By Theorem 3.1, there exists a continuous injective curve $\gamma : [0, 1] \to C$ passing through every sample point. A learner minimising samplewise loss can assign each x_i a one-dimensional coordinate and learn a readout that achieves perfect training accuracy using only this traversal coordinate.

The traversal representation has no concept of the wrap-around at $\theta = 2\pi \equiv 0$, and no encoding of the fact that phases $\theta = 0.01$ and $\theta = 2\pi - 0.01$ are semantically adjacent.

> **Failure Mode 1:** The representation is consistent with the data but inconsistent with the quotient. Any evaluation that depends only on samplewise accuracy will report success. The circular structure of S^1 is invisible to such an evaluation.

3.4 Why Density Does Not Help: The N-Turn Spiral

Define the N-turn spiral

$$\gamma_N(s) = \big(\cos(2\pi N s),\, \sin(2\pi N s),\, s\big), \quad s \in [0, 1].$$

As $N \to \infty$ its image becomes uniformly dense on the cylinder. Yet the image of γ_N is homeomorphic to a closed interval for every N: its local neighbourhoods are one-dimensional line segments. The spiral never embeds the cylinder—it merely covers it densely.

> Geometric approximation error and topological faithfulness are independent. A 1D representation can achieve reconstruction error below any desired threshold on any finite sample while remaining permanently, provably unable to represent the quotient faithfully.

This sharpens the empirical finding of Zhang et al. [2017] that deep networks can fit arbitrary label assignments. Theorem 3.1 gives the geometric reason. The distinction between "SGD finds some structure" and "SGD finds the right structure" is exactly the gap this paper addresses.

3.5 From Ambiguity to Admissibility

Theorem 3.1 and Corollary 3.3 establish one negative fact: finite predictive fit cannot certify the intrinsic structure of the generating process. The finite-sample traversal result and the sufficiency collapse result of Section 4 are the same phenomenon at different levels of analysis.

4 Sufficiency and Compression Do Not Certify Lawful Abstraction

One might hope that minimal sufficient compression would close the gap identified in Section 3. This section shows that this hope is false, in a precise and general sense. The minimal sufficient statistic for a prediction target is, whenever the task map is non-injective, a structurally wrong object with respect to the lawful quotient: the global minimum of compression sits at a point that collapses $\mathcal{Q}$.

4.1 Definitions

Definition 4.1 (Sufficient representation)**.** A representation $E : X \to Z$ is *sufficient* for f if there exists a measurable map $g : Z \to Y$ such that $f = g \circ E$.
Definition 4.2 (Compression preorder)**.** For representations $E_1 : X \to Z_1$ and $E_2 : X \to Z_2$, write $E_1 \preceq E_2$ if there exists a measurable map $h : Z_2 \to Z_1$ such that $E_1 = h \circ E_2$. We say E_1 is *at least as compressed as* E_2.
Definition 4.3 (Minimal sufficient representation)**.** A representation E is *minimal sufficient* for f if it is sufficient for f and $E \preceq E'$ for every sufficient representation E'.

4.2 The Main Theorem

Theorem 4.4 (Task-Map Collapse)**.** *Parts (i) and (ii) restate, in the compression-preorder framework, the classical result that the label function is the minimal sufficient statistic [Lehmann and Scheffé, 1950, Tishby et al., 1999]. Part (iii) connects injectivity of the task map to topological embeddability of the quotient.*

Define the label representation by $E_{\text{label}}(x) = f(x)$. Then:

(i) E_{label} is sufficient for f.

(ii) E_{label} *is the minimum of the compression preorder restricted to sufficient representations:* $E_{\text{label}} \preceq E$ *for every sufficient* E*.*

(iii) Unless $t : \mathcal{Q} \to Y$ *is injective,* E_{label} *does not embed* $\mathcal{Q}$*. Specifically, the map induced on the quotient is the task map* t *itself, and if* t *identifies two distinct quotient classes then* E_{label} *cannot be a topological embedding of* $\mathcal{Q}$*.*

Proof. See Appendix B. □

Remark 4.5 (Uniqueness of the minimum). The minimum element E_{label} is unique in the compression preorder among sufficient representations: any two representations that are each minimal must be equivalent up to relabelling of Z, since each factors through the other. The minimum is the unique canonical element at the bottom of the sufficiency order, determined entirely by the task map f.

A Kolmogorov complexity interpretation of why E_{label} is specifically the minimum—rather than some other sufficient representation—is developed in Paper 2 [Farkas, 2026b], Remark 2.2a: $E_{\text{label}} = t \circ q$ is the KC-simplest sufficient encoder, and simplicity bias in deep networks makes it the preferred solution under any simple task-loss objective. This paper establishes the structural fact; the complexity-theoretic explanation is in Paper 2.

4.3 Illustration: Over-Compression on the Phase Cylinder

The declared quotient is $\mathcal{Q} = S^1$ and the latent space is $Z = \mathbb{R}^2$, which can in principle embed S^1 faithfully. Suppose the learner optimises for the most compressed sufficient statistic for the sine-projection target $t(\theta) = \sin(\theta)$.

By Theorem 4.4, $E_{\text{label}}(x) = \sin(\theta(x))$ is the minimum of the compression preorder. It assigns the same code to every pair $(\theta, \pi - \theta)$. The image of S^1 under t is the interval $[-1, 1]$, not a circle.

> **Failure Mode 2:** The representation satisfies sufficiency, quotient respect, and is optimal by every compression criterion—but fails structural faithfulness. The collapse of S^1 onto $[-1, 1]$ is invisible to regression accuracy.

4.4 The Separation: Target Sufficiency vs. Quotient Faithfulness

Minimal sufficiency answers a question about Y: what is the most compressed representation that still permits perfect prediction of f? Quotient faithfulness answers a question about $\mathcal{Q}$: does the representation preserve the structure of the declared semantic space? They coincide only when t is injective. When t is not injective, the global optimum of the first collapses the second: the loss landscape places its minimum at a point that fails structural faithfulness.

Note carefully that E_{label} *does* respect the quotient: it assigns the same code to all observations in the same nuisance fiber. The failure is not that it violates the nuisance structure; the failure is that the induced map on $\mathcal{Q}$ is too coarse. Minimal sufficient compression has collapsed $\mathcal{Q}$ further than the nuisance collapse prescribed by q. This is the precise sense in which sufficiency does not certify faithfulness.

4.5 Contrast with Information-Bottleneck Objectives

The information bottleneck [Tishby et al., 1999, Tishby and Zaslavsky, 2015] proposes to learn representations by minimising $I(X; Z)$ subject to preserving $I(Z; Y)$. At optimality

this produces the minimal sufficient statistic of X for Y—precisely the object that Theorem 4.4 shows need not embed the lawful quotient.

The same phenomenon appears in modern self-supervised learning. Tian et al. [2020] formalise this: the choice of augmentation controls which equivalences the representation enforces. When the augmentation-defined task map is non-injective on $\mathcal{Q}$, the global optimum of the contrastive objective fails to embed $\mathcal{Q}$ regardless of model capacity. This is not a failure of expressive power but of objective structure: the IB optimum is E_{label} precisely because it is the KC-simplest sufficient encoder [Farkas, 2026b, Remark 2.2a], and the IB objective rewards that simplicity.

5 Structural Components of Lawful Abstraction

5.1 The Three Requirements

Proposition 5.1 (Lawful Abstraction Decomposition)**.** *For a representation $E : X \to Z$, the following three properties are logically distinct. No one of them implies the conjunction of the other two.*

(I) ***Task sufficiency.*** *There exists a measurable map $g : Z \to Y$ such that $f = g \circ E$.*
(II) ***Quotient respect.*** *There exists a measurable map $\tilde{E} : \mathcal{Q} \to Z$ such that $E = \tilde{E} \circ q$.*
(III) ***Structural faithfulness.*** *The induced map $\tilde{E} : \mathcal{Q} \to Z$ is a topological embedding of $\mathcal{Q}$. In stronger settings, $\tilde{E}$ is additionally required to be bi-Lipschitz.*

A lawful abstraction *is a representation satisfying all three.*

The three properties are logically independent: no two imply the third. The label representation $E_{\text{label}} = t \circ q$ satisfies (I) and (II) while failing (III) whenever t is non-injective. A topological embedding $\tilde{E}$ composed with q satisfies (II) and (III) while potentially failing (I). A sufficient representation that retains nuisance alongside predictive features satisfies (I) while failing (II).

5.2 Relation to Disentanglement

The central result of Locatello et al. [2019] is that unsupervised disentanglement is impossible without inductive biases: the correct factorisation cannot be identified from data alone. However, disentanglement is not the same as quotient faithfulness. A perfectly disentangled representation may still fail requirement (III).

The precise difference is this: the Locatello et al. impossibility is an identification problem. The impossibility established by Theorem 4.4 operates downstream of that problem: even after the correct quotient $\mathcal{Q}$ has been declared externally, compression objectives converge to a representation that fails requirement (III). The two results are logically independent and jointly exhaustive of the failure space.

6 From Lawful Quotients to Embedding Constraints

Section 4 showed one side of the structural problem: compression pulls the representation toward the label space. The topology and metric results below show the other side: even with the quotient fixed and respected, the latent space may be geometrically incapable of representing it faithfully.

6.1 What This Reduction Does and Does Not Say

Classical universal approximation results—Cybenko [1989], Hornik [1991]—guarantee that for any continuous target function, a sufficiently expressive network can approximate it in value. A network can be a universal function approximator and still induce a $\tilde{E}$ that is not injective, not continuous, or not a homeomorphism. Value universality and topology universality are independent properties.

7 Embedding Obstructions to Topology-Faithful Abstraction

7.1 Topology-Faithful Representations

Definition 7.1 (Topology-faithful representation)**.** A continuous map $\tilde{E} : \mathcal{Q} \to Z$ is *topology-faithful* if it is a topological embedding: a homeomorphism from $\mathcal{Q}$ onto its image $\tilde{E}(\mathcal{Q}) \subset Z$ equipped with the subspace topology.

Homeomorphism is the right criterion because it preserves all continuous topological invariants: connectedness, compactness, and local neighbourhood structure.

7.2 The Obstruction Theorems

Theorem 7.2 (Quotient Embedding Requirement; Menger 1932, Kvalheim and Sontag 2024)**.** *Let $\mathcal{Q}$ be a compact Hausdorff space and let $\tilde{E} : \mathcal{Q} \to \mathbb{R}^m$ be continuous. If $\tilde{E}$ is topology-faithful, then $\mathcal{Q}$ admits a topological embedding into $\mathbb{R}^m$.*

Equivalently: if $\mathcal{Q}$ is not embeddable in $\mathbb{R}^m$, then no quotient-respecting representation with exposed latent space $\mathbb{R}^m$ can be topology-faithful.

Proof. Definitional. See Appendix B. □

Remark 7.3 (Relationship to Kvalheim–Sontag (2024))*.* It is important to be transparent about the mathematical status of this theorem. Theorem 7.2 is logically equivalent to the definition of a topological embedding and has no content beyond it. Furthermore, Kvalheim and Sontag [2024] have already proved the exact autoencoder-specific version: a perfect autoencoder with bottleneck $\mathbb{R}^k$ exists if and only if the data manifold is homeomorphic to a subset of $\mathbb{R}^k$, which directly subsumes the mathematical strength of Theorem 7.2. We restate it here not as a novel mathematical contribution, but because it forms the third and final necessary pillar of the diagnostic framework: the obstruction is on the target space, not the computation, and this needs to be stated explicitly within the three-failure-mode structure. The genuinely new content in this section is the application to the compression preorder framework and the connection to Failure Mode 3.

Theorem 7.4 (Embedding-Dimension Obstruction; Brouwer 1911, Hurewicz and Wallman 1941, Engelking 1978)**.** *Let $\mathcal{Q}$ be a compact metric space of finite covering dimension k, and let $\tilde{E} : \mathcal{Q} \to \mathbb{R}^m$ be topology-faithful. Then $k \leq m$.*

Proof. Direct consequence of monotonicity of covering dimension for subspaces of Euclidean space. See Appendix B. □

The $m = 1$ case recovers the scalar bottleneck result: the only compact connected spaces faithfully encodable by a scalar are closed intervals. Any quotient with nontrivial loop

structure—S^1, S^k, T^k, any compact manifold of dimension ≥ 2—requires at least a two-dimensional latent space.

Corollary 7.5 (Below-Embedding Collapse)**.** *Let $\mathcal{Q}$ be a compact metric space with covering dimension k, and let $\tilde{E} : \mathcal{Q} \to \mathbb{R}^m$ with $m < k$. Then $\tilde{E}$ cannot be topology-faithful. Any such map must either identify distinct quotient classes, fail to preserve neighbourhood topology, or both.*

Proof. Immediate contrapositive of Theorem 7.4. See Appendix B. □

Remark 7.6 (Entanglement, not destruction). Corollary 7.5 establishes that no topology-faithful encoding exists when $\dim(Z) < \dim(Q)$. It does not say that the quotient structure is absent from the representation. A continuous map $\tilde{E} : Q \to Z$ with $\dim(Z) < \mathrm{emb}(Q)$ still exists and will be produced by any trained encoder; what is impossible is for that map to be injective in a topologically stable way.

What actually occurs is *entanglement*: the encoder folds Q into the available dimensions, creating false identifications between semantically distinct states. Distinct points of Q must be collapsed together (the map is non-injective), and the information about which quotient class a point belongs to becomes mixed with whatever else occupies the dimensions of Z. The quotient structure is not erased — it is co-mingled with other encoded content in a way that makes it non-independently recoverable without simultaneously decoding the other entangled dimensions.

This is a different failure mode from the compression trap (Theorem 4.4), where the objective actively rewards collapsing Q. The dimension trap is a geometric impossibility that operates regardless of the objective: even a perfect objective cannot produce a topology-faithful map from Q into a space too small to embed it. The compression trap destroys distinctions because the task rewards it; the dimension trap entangles distinctions because the geometry requires it.

7.3 Why the Dimension Lower Bound Is a Floor, Not a Practical Design Target

Theorem 7.4 gives a lower bound: $\dim(Z) \geq \dim(\mathcal{Q})$ is necessary for topology-faithful representation. In practice this bound is very loose.

Frankle and Carlin [2019] showed that the winning lottery ticket subnetworks only train successfully when reset to their original initialization. This means the structural information learned during training is entangled with specific weight values—it is not stored in the topology of the subgraph alone.

A plausible implication for latent-space design is that networks may use surplus dimensions of Z to distribute and stabilise learned quotient structure across many weights. The dimension lower bound of Theorem 7.4 is therefore a floor, not a target: in practice, architectures may require substantially more capacity than $\dim(Z) \geq \dim(\mathcal{Q})$ when the learned structure is distributed rather than cleanly localized.

> This suggests that in practice $\dim(Z)$ may need to exceed $\dim(\mathcal{Q})$ substantially when the learned structure is distributed rather than cleanly localized.

Remark 7.7 (The bound is on the exposed representation, not on layer widths). Theorem 7.4 constrains the *exposed latent space* Z—the representation produced by the encoder as a whole, before any prediction head. It says nothing about the dimension of any individual layer, intermediate activation, or architectural bottleneck inside the network. A deep

network with a narrow internal layer can produce an exposed representation in $\mathbb{R}^m$ for any m; the theorem requires that $m \geq \dim(\mathcal{Q})$ but is silent on how the network achieves this internally.

In particular, the theorem should *not* be read as implying that every layer must have width at least $\dim(\mathcal{Q})$, or that a network with any layer of width less than $\dim(\mathcal{Q})$ cannot produce a topology-faithful exposed representation. The connection between internal architecture and the structural properties of the exposed representation is a realizability question addressed in the forthcoming follow-up paper (Farkas, in preparation). The present results are purely about what the exposed map $\tilde{E} : \mathcal{Q} \to Z$ can and cannot achieve as a function, independent of how the encoder is implemented.

Hofer et al. [2017] introduced differentiable topology layers that penalise deviations from a target topological structure. Moor et al. [2023] proposed the Representation Topology Divergence (RTD). Both methods directly operationalise Definition 7.1.

These methods confirm the practical relevance of the topological constraints proved here. Constructive scalable approaches to enforcing quotient structure are outside the scope of this paper and are addressed in the follow-up.

7.4 Metric Distortion: Beyond Topology-Faithfulness

Topology-faithfulness is necessary but not sufficient for downstream reasoning. A homeomorphism can distort distances arbitrarily while remaining valid throughout.

7.4.1 A Family of Topology-Faithful Maps with Unbounded Distortion

Proposition 7.8 (Topology–Geometry Gap; cf. Burago et al. 2001)**.** *Let $\mathcal{Q} = [0, 1]$ with the standard metric. For $\alpha \geq 1$ define $\tilde{E}_\alpha(x) = x^\alpha$. Each $\tilde{E}_\alpha$ is a homeomorphism of $[0, 1]$ onto itself, hence topology-faithful. However, the metric distortion of $\tilde{E}_\alpha$ is unbounded: as $\alpha \to \infty$, neighbourhoods of fixed size near 0 collapse to arbitrarily small images.*

Proof. See Appendix B. □

> Nearest-neighbour retrieval, clustering, interpolation between latent codes, and any other operation that treats distances in Z as proxies for semantic similarity all depend on the metric structure of Z approximating the metric structure of $\mathcal{Q}$.

7.4.2 Metric-Faithful Representations

Definition 7.9 (Metric-faithful representation)**.** A continuous map $\tilde{E} : \mathcal{Q} \to \mathbb{R}^m$ is *metric-faithful* if it is bi-Lipschitz: there exist constants $0 < c \leq C < \infty$ such that

$$c \cdot d_{\mathcal{Q}}(q_1, q_2) \leq \|\tilde{E}(q_1) - \tilde{E}(q_2)\| \leq C \cdot d_{\mathcal{Q}}(q_1, q_2)$$

for all $q_1, q_2 \in \mathcal{Q}$.

Remark 7.10 (The hierarchy)*.* The structural requirements on $\tilde{E}$ form a strict hierarchy:

$$\text{metric-faithful} \implies \text{topology-faithful} \implies \text{continuous} \implies \text{measurable}.$$

Each arrow is strict. For lawful abstraction, the minimum required is topology-faithful. For operations that rely on distance-based reasoning in Z, metric-faithful is required.

7.4.3 Where Metric Distortion Arises in Practice

Wang and Isola [2020] show that the contrastive loss asymptotically optimises alignment—collapsing nuisance-equivalent observations to nearby codes—and uniformity—preventing concentration of large regions of $\mathcal{Q}$ onto small regions of Z. In the language of this paper, uniformity is a soft form of the lower Lipschitz condition.

A concrete mechanism for near-manifold metric distortion arises in diffusion models. Sahraee-Ardakan et al. [2026] identify the Jensen Gap: a model trained to predict $\mathbb{E}[\varepsilon|u]$, marginalised over noise levels, systematically diverges from the true score $\mathbb{E}[\varepsilon|u,t]$. In the language of this paper, this is a specific instance of Failure Mode 3: the representation may be topology-faithful but metrically distorted near the data manifold.

7.4.4 Illustration: Metric Distortion on the Phase Cylinder

Suppose the learner applies a topology-enforcing objective that detects the missing 1-cycle and closes the representation into a loop. Gradient descent closes the loop by collapsing consecutive turns at phases θ and $\theta + 2\pi/N$, producing a closed curve that is a topological embedding of S^1. The topology loss reports success.

The metric geometry is wrong. The bi-Lipschitz lower constant $\alpha \to 0$ as $N \to \infty$.

> **Failure Mode 3:** The representation satisfies sufficiency, quotient respect, and topology-faithfulness—but fails metric-faithfulness. The topology loss has done exactly what it was asked to do. What it was not asked to do—and what no standard training objective asks for—is to distribute arc length faithfully.

7.4.5 Extension: The Cylinder Quotient and Dimension

The cylinder quotient $\mathcal{Q} = S^1 \times [0,1]$ is two-dimensional. By Theorem 7.4, any topology-faithful representation requires $\dim(Z) \geq 2$. No continuous map from the two-dimensional cylinder into a one-dimensional space can be a topological embedding.

Summary

Remark 7.11 (Landscape structure vs. training dynamics). The three results of Sections 3–7 are claims about the structure of the loss landscape and the certification problem, not about training trajectories.

Corollary 3.3 establishes that the samplewise objective admits a perfect-loss solution consistent with a one-dimensional traversal of the sample cloud. It does not claim that gradient descent will find such a traversal, or that no initialisation or parameterisation can avoid it.

Theorem 4.4 establishes that the global minimum of the compression objective is a structurally wrong point: the minimal sufficient statistic collapses $\mathcal{Q}$ whenever t is non-injective. It does not claim that training always converges to this minimum. Whether gradient descent reaches it, avoids it, or settles near it depends on the basin of attraction, the initialisation, the architecture, and the recovery mechanisms studied in the companion paper (Farkas, in preparation).

Theorems 7.2 and 7.4, and Corollary 7.5, establish geometric impossibility conditions on $\tilde{E} : \mathcal{Q} \to Z$. These are unconditional: no continuous map from $\mathcal{Q}$ into an insufficiently

large Z can be topology-faithful, regardless of training procedure. They do not assert that any particular training procedure will induce such an under-dimensional map.

The correct reading of this paper is therefore: *these are the structural features of the loss landscape that make standard criteria insufficient for certifying quotient faithfulness.* The recovery question—when and how a system can escape the traversal and compression traps despite what the landscape permits—is what the companion paper addresses. Paper I establishes why recovery mechanisms are needed; the companion paper characterises what they must do.

This paper has established three logically independent reasons why standard training criteria cannot certify quotient faithfulness. The Traversal Ambiguity Corollary shows that any finite sample is consistent with a one-dimensional interpolation, so training accuracy provides no evidence that intrinsic quotient structure has been recovered. The Task-Map Collapse Theorem—the central result—shows that the global optimum of the compression objective is a structurally wrong point whenever the task map is non-injective, which is the common case: the minimal sufficient statistic collapses quotient structure, and no task-based criterion can detect this. The Embedding Obstruction Theorems show that faithful encoding imposes hard geometric constraints on the latent space that no prediction objective enforces: the latent space must embed the quotient $\mathcal{Q}$ both topologically and metrically, and these requirements follow from the geometry of $\mathcal{Q}$ itself and apply to any induced map $\tilde{E} : \mathcal{Q} \to Z$ once the latent space Z is fixed. While the strict geometric constraints are grounded in the established work of Brouwer (1911), Hurewicz–Wallman (1941), and—in the ML context—Kvalheim and Sontag [2024], the contribution here is their integration into the three-failure-mode diagnostic structure alongside the sufficiency and traversal results.

Supervised losses, contrastive objectives, and information-bottleneck variants do not, by themselves, certify quotient structure—the theorems explain why. Architectural commitments such as equivariant networks, JEPA-style latent prediction, and diffusion models partially escape the collapse by baking structural priors into the architecture rather than the loss. The present results characterise precisely what is left uncertified even in those cases.

The compression preorder and the quotient faithfulness criterion are the paper's two formal additions. They give precise language for a gap that the representation learning literature has approached informally: the difference between a representation that is optimal for prediction and one that is faithful to the structure of the data-generating process. Theorem 4.4(iii) is the hinge—it identifies the minimal sufficient statistic not as an approximation to the ideal representation but as the object that provably collapses the quotient whenever the task map is non-injective.

What this paper does not address is how quotient faithfulness can be achieved. The gap it identifies is real but not permanent. When admissibility structure is known, architectures can be designed to respect it; how stochastic mechanisms may assist quotient discovery when admissibility is only partially specified is left to the follow-up paper. Those constructive questions—recovery conditions, architecture complexity bounds, and stochastic escape mechanisms—are the subject of the forthcoming follow-up paper (Farkas, in preparation).

The present paper establishes a limitation of certification, not a limitation of empirical success. A network may still recover a lawful quotient in practice, but when it does, that

success must come from structure not captured by the criteria analysed here. The natural next question is therefore constructive rather than negative: which additional biases, objectives, and architectural constraints make quotient recovery possible, and under what conditions can they be interpreted as admissibility signals rather than mere heuristics? Those questions are left to the forthcoming follow-up paper (Farkas, in preparation).

Appendix B contains proofs of classical results; Appendix C gives the theorem-by-theorem attribution; Appendix D gives the AI disclosure. The first author takes full scientific responsibility for all claims.

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A Definitions and Key Concepts

This appendix collects the paper's formal definitions for quick reference.

Admissibility and quotient structure.

Admissible nuisance relation (§2.1). An equivalence relation $\sim$ on $\mathcal{X}$ identifying observations that differ only in semantically irrelevant ways.

Lawful quotient (§2.2). (Q, q) where $Q = \mathcal{X}/\sim$ and $q : \mathcal{X} \to Q$ is the canonical projection. Q is the space of semantic states.

Representation map / encoder (§2.4). A measurable map $E : \mathcal{X} \to Z$ into the exposed latent space Z.

Quotient-respecting representation (§2.5). E factors as $E = \tilde{E} \circ q$; equivalently, $E(x) = E(x')$ whenever $x \sim x'$.

Compression and sufficiency.

Sufficient representation (§4.1). E is sufficient for f if $f = g \circ E$ for some measurable g.

Compression preorder (§4.1). $E_1 \preceq E_2$ iff $E_1 = h \circ E_2$ for some measurable h.

Minimal sufficient representation (§4.1). E is sufficient and $E \preceq E'$ for every sufficient E'.

Label representation (§4.2). $E_{\text{label}}(x) = f(x)$. The unique $\preceq$-minimum among sufficient representations; collapses Q whenever t is non-injective.

Faithfulness conditions.

Topology-faithful representation (§7.1). $\tilde{E} : Q \to Z$ is a topological embedding: a homeomorphism from Q onto $\tilde{E}(Q) \subset Z$.

Metric-faithful representation (§7.4.2). $\tilde{E}$ is bi-Lipschitz with constants $0 < c \leq C < \infty$.

Lawful abstraction (§5.1). A representation satisfying task sufficiency, quotient respect, and structural faithfulness simultaneously.

B Proofs of Classical Results

This appendix collects the proofs for all results in the main text that apply classical mathematics without new ideas. None of the proofs in this appendix constitute a mathematical contribution of this paper.

B.1 Proof of Theorem 3.1 (Finite-Sample Traversal)

Classical source: Whyburn (1942); Rourke–Sanderson (1972), Ch. 1.

Proof. Because the points are pairwise distinct, there exists a vector $v \in \mathbb{R}^m$ such that the projections $\lambda_i = \langle v, x_i \rangle$ are pairwise distinct. The set of bad directions—those v for which some pair of projections coincide—is a finite union of proper hyperplanes $\{v : \langle v, x_i - x_j \rangle = 0\}$, which has Lebesgue measure zero; a generic v avoids all of them.

Relabel so that $\lambda_1 < \lambda_2 < \cdots < \lambda_n$. Define γ piecewise linearly on $[(i-1)/(n-1),\ i/(n-1)]$ by

$$\gamma\left(\frac{i-1+s}{n-1}\right) = (1-s) \cdot x_i + s \cdot x_{i+1}, \quad s \in [0,1].$$

This map is continuous by construction and passes through each x_i. For injectivity, consider $\phi(t) = \langle v, \gamma(t) \rangle$. On the i-th segment,

$$\phi\left(\frac{i-1+s}{n-1}\right) = (1-s)\lambda_i + s\lambda_{i+1},$$

which is strictly increasing in s because $\lambda_i < \lambda_{i+1}$. Hence ϕ is strictly increasing on $[0,1]$, so $t \neq t'$ implies $\gamma(t) \neq \gamma(t')$. □

B.2 Proof of Corollary 3.3 (Traversal Ambiguity)

Proof. Apply Theorem 3.1 to obtain a continuous injective $\gamma : [0,1] \to \mathbb{R}^m$ with $\gamma(t_i) = x_i$. Set $\tau(x_i) = t_i$ and choose h so that $h(t_i) = y_i$ for all i, e.g. by piecewise linear interpolation for scalar targets. □

B.3 Proof of Theorem 4.4 (Task-Map Collapse)

Classical sources for (i)–(ii): Lehmann–Scheffé (1950); Shamir, Sabato, and Tishby (2010).

Proof. **(i) Sufficiency.** By definition $E_{\text{label}}(x) = f(x)$, so the identity map recovers the target: $f = \text{id} \circ E_{\text{label}}$.

(ii) Minimality. Let $E : X \to Z$ be any sufficient representation. Then there exists $g : Z \to Y$ with $f = g \circ E$. Since $E_{\text{label}}(x) = f(x)$ for all x, substituting gives $E_{\text{label}} = g \circ E$, which is exactly $E_{\text{label}} \preceq E$ by definition. Since E was arbitrary, E_{label} is a lower bound of all sufficient representations. Because it is itself sufficient, it is the minimum element.

(iii) Failure to embed. Because $f = t \circ q$, the label representation factors as $E_{\text{label}} = t \circ q$, so the induced map on $\mathcal{Q}$ is $\tilde{E}_{\text{label}} = t$. If t is not injective, there exist distinct $q_1, q_2 \in \mathcal{Q}$ with $t(q_1) = t(q_2)$, giving $\tilde{E}_{\text{label}}(q_1) = \tilde{E}_{\text{label}}(q_2)$. A topological embedding must be injective, so $\tilde{E}_{\text{label}}$ is not a topological embedding of $\mathcal{Q}$. □

B.4 Proof of Theorem 7.2 (Quotient Embedding Requirement)

Proof. If $\tilde{E}$ is topology-faithful, then by Definition 7.1 it is a homeomorphism from $\mathcal{Q}$ onto $\tilde{E}(\mathcal{Q}) \subset \mathbb{R}^m$. A homeomorphism onto a subspace is by definition a topological embedding. The contrapositive gives: if $\mathcal{Q}$ is not embeddable in $\mathbb{R}^m$, no topology-faithful map $\mathcal{Q} \to \mathbb{R}^m$ exists. □

B.5 Proof of Theorem 7.4 (Embedding-Dimension Obstruction)

Classical sources: Brouwer (1911); Hurewicz–Wallman (1941); Engelking (1978), Theorem 3.1.2.

Proof. Because $\tilde{E}$ is topology-faithful, $\mathcal{Q}$ is homeomorphic to $\tilde{E}(\mathcal{Q}) \subset \mathbb{R}^m$. Covering dimension is a topological invariant (Engelking 1978, Theorem 1.7.7): homeomorphic spaces have equal covering dimension. It is also monotone for subspaces of metrizable spaces (Engelking 1978, Theorem 3.1.2): any subspace of $\mathbb{R}^m$ has covering dimension at most m. Therefore

$$k = \dim(\mathcal{Q}) = \dim(\tilde{E}(\mathcal{Q})) \leq \dim(\mathbb{R}^m) = m. \qquad \square$$

Remark B.1 ($m = 1$, scalar bottleneck). When $m = 1$, the theorem gives $k \leq 1$ for any faithfully encoded compact connected quotient $\mathcal{Q}$. Combined with Munkres (2000), Theorems 24.1 and 27.3—which characterise compact connected subsets of $\mathbb{R}$ as closed intervals—this gives $\mathcal{Q} \cong [a, b]$. Any quotient with nontrivial loop structure (S^1, T^k, S^k for $k \geq 2$) cannot be faithfully encoded by a scalar.

B.6 Proof of Corollary 7.5 (Below-Embedding Collapse)

Proof. Immediate contrapositive of Theorem 7.4: if a topology-faithful map existed, we would have $k \leq m$, contradicting $m < k$. □

B.7 Proof of Proposition 7.8 (Topology–Geometry Gap)

Classical source: Burago, Burago, and Ivanov (2001), Chapter 1.

Proof. **Topology-faithfulness.** Each $\tilde{E}_\alpha$ is strictly increasing and continuous with continuous inverse $x \mapsto x^{1/\alpha}$, so it is a homeomorphism of $[0, 1]$ onto itself.

Unbounded distortion. Take $x_\alpha = 0$, $y_\alpha = 1/\alpha$. The source distance is $d_{\mathcal{Q}}(x_\alpha, y_\alpha) = 1/\alpha$. The image distance is

$$|\tilde{E}_\alpha(x_\alpha) - \tilde{E}_\alpha(y_\alpha)| = (1/\alpha)^\alpha = \alpha^{-\alpha}.$$

The compression ratio is $\alpha^{-\alpha}/(1/\alpha) = \alpha^{1-\alpha} \to 0$ as $\alpha \to \infty$. □

C Novelty and Contribution Attribution

This appendix provides a unified account of (i) the novelty status of each formal result, and (ii) the intellectual contribution of each party to the paper. The two assessments are presented together because they address the same underlying question: what is genuinely new here, and who produced it?

The attribution table below uses the weight scheme from Appendix D: **0** (not applicable), + (substantive contribution), ++ (primary responsibility; this party could have produced this result independently). The **AI** ++ weight is reserved for results volunteered without being asked; executing a given direction is +, not ++.

Attribution Table

Result / Contribution	Character	Lit.	Author	AI
Framework and formalisations				
Three-failure-mode diagnostic structure	novel framework	0	++	0
Compression preorder (Definition 4.2)	novel formalisation	0	++	0
Quotient faithfulness criterion (Definition 7.1)	novel formalisation	0	++	0
Lawful Abstraction Decomposition (Prop. 5.1)	novel framing	0	++	0
Cylinder / phase / sine running example	novel illustration	0	++	0
Theorems applied as tools from classical mathematics				
Thm 3.1 Finite-Sample Traversal	classical folklore	++	+	0
Cor 3.3 Traversal Ambiguity (ML framing)	classical consequence	+	++	0
Thm 4.4 (i–ii) Minimal Sufficiency	classical statistics	++	+	0
Thm 7.2 Quotient Embedding Requirement	tautological reframe	++	0	0
Thm 7.4 Embedding-Dimension Obstruction	classical dim. theory	++	+	0
Cor 7.5 Below-Embedding Collapse	immediate consequence	++	0	0
Prop 7.8 Topology–Geometry Gap	classical metric geom.	++	+	0
Novel connections and observations				
Thm 4.4 (iii) Task-Map Collapse	genuinely new	+	++	0
Global loss minimum traversal-achievable (Rem. 3.3.1)	novel observation	0	++	0
Traversal–universal approximation connection (§6.1)	novel synthesis	+	++	0
K&S subsumption acknowledgement (Rem. 7.3)	novel framing	+	++	0
Scientific direction and editorial decisions				
Decision to replace AI proofs with classical citations	scientific direction	0	++	0
Novelty classification of all theorems	scientific direction	0	++	0
Reference selection with interpretive framing	scientific direction	0	++	0
Recent additions (this revision, Apr. 2026)				
KC forward reference in Rem. 4.5 (uniqueness of minimum)	connecting note	+	++	+
KC sentence in §4.5 (IB objectives)	connecting note	+	++	+
Notation page (§*Notation)	structural addition	0	++	+
Definitions appendix (App. A)	structural addition	0	++	+
AI-assisted execution (all under author direction)				
Source proof drafts (prior to author substitution pass)	execution	0	0	+
Prose drafting from author outlines and notes	execution	0	0	+
LaTeX typesetting of author definitions	execution	0	0	+
Literature search and citation formatting	execution	0	0	+

C.1 Primary Contribution

The paper's primary contribution is a unified diagnostic framework that proves three logically independent reasons why standard training criteria cannot certify quotient faithfulness, together with two novel formalisations—the compression preorder and the quotient faithfulness criterion—that give precise language for the gap. The individual mathematical theorems are classical results applied in a new context. The genuinely novel intellectual content is the framework itself and the connections it draws between classical topology, statistical sufficiency, and representation learning evaluation.

C.2 Critical Prior Work

The three most important prior works are: Locatello et al. [2019], whose unidentifiability result for disentanglement is the closest structural predecessor to the Task-Map Collapse theorem; Tishby et al. [1999], Tishby and Zaslavsky [2015], whose information bottleneck is the specific compression objective shown to actively destroy quotient structure; and Kvalheim and Sontag [2024], whose topological results on autoencoders are strictly stronger than Theorem 7.2 and must be engaged prominently.

C.3 AI Tools

Tool	Version	Phase	Role
ChatGPT	GPT-4 era	Ideation	Exploratory conversation partner
GPT-5.4	OpenAI, Mar. 2026	Proof drafts	Source proof drafts (superseded by author)
Claude Sonnet 4.6	Anthropic, Apr. 2026	Writing	Prose drafting; KC forward references; LaTeX typesetting; all under author direction

Generative AI outputs are non-deterministic and cannot be exactly reproduced. The paper should be evaluated on its intellectual and mathematical content alone. The intellectual genesis of all novel contributions belongs to the first author; the AI's role was expressive, not generative. The first author takes full scientific and intellectual responsibility for all claims.

D The Attribution Prompt and Worked Example

D.1 The Prompt

The following prompt can be used to generate a contribution attribution table for any AI-assisted academic paper. It is designed to be honest and reproducible.

> Read the full paper and the full conversation history before making any attributions. The paper alone cannot establish who originated an idea or whether the AI acted on a direction or volunteered something unprompted. The conversation history is the primary evidence for Author vs. AI weight assignments. Attributions made without it are unreliable.

Identify every distinct intellectual unit: named theorems, definitions, conjectures, remarks, open problems, syntheses of existing results, reframings of existing phenomena, and novel connections. Each is a separate row. Do not include purely pedagogical illustrations.

For each row, write a short descriptive label in the Character column—one to four words, honest and descriptive. A row that is genuinely two things can say so.

Assign weights across three columns—**Lit.** (prior literature), **Author**, **AI**—using:

- **0** not applicable
- **+** substantive contribution
- **++** primary responsibility; this party could have produced this result independently

Calibration: Author ++ when the idea originated with the human author and would not have appeared without their specific input. AI ++ reserved exclusively for results volunteered without being asked—not for executing a given direction, not for writing prose, not for identifying a theorem when prompted. Supplying a theorem when asked is +. Proposing its relevance unprompted is ++.

Produce the table, then write: a **Primary Contribution** paragraph stating the paper's main contribution plainly; a **Critical Prior Work** subsection naming the two or three papers the work builds on most directly; a check for any conflation between existence results and constructive or learnability results; and an **AI Tools** subsection listing each tool, its version, phase, and role, closing with a statement that the human author takes full responsibility for all claims.

D.2 Worked Example: This Paper

The table below was produced by applying the prompt above to Paper 1 after reading the full drafting conversation. The AI ++ column is empty throughout: no row in this paper meets the criterion of an unsolicited contribution that the AI volunteered without being asked. Every AI contribution was in response to a specific direction from the author. This is the correct result for a paper in which the author provided detailed mathematical outlines, directed the proof strategy, selected all references, and made the decision to replace AI-generated proofs with classical citations.

Result / Contribution	Character	Lit.	Author	AI
Three-failure-mode framework	novel framework	0	++	0
Compression preorder	novel formalisation	0	++	0
Quotient faithfulness criterion	novel formalisation	0	++	0
Lawful Abstraction Decomposition	novel framing	0	++	0
Running example (cylinder/phase/sine)	novel illustration	0	++	0
Thm 3.1 Traversal	classical	++	+	0
Cor 3.3 Traversal Ambiguity	ML framing	+	++	0
Thm 4.4 (i–ii) Minimal Sufficiency	classical	++	+	0
Thm 4.4 (iii) Task-Map Collapse	genuinely new	+	++	0
Thm 7.2 Embedding Requirement	tautological	++	0	0
Thm 7.4 Dimension Obstruction	classical	++	+	0
Cor 7.5 Below-Embedding Collapse	consequence	++	0	0
Global minimum / traversal link	novel observation	0	++	0
Traversal–UAT connection	novel synthesis	+	++	0
K&S subsumption acknowledgement	novel framing	+	++	0
Classical-source substitution decision	scientific direction	0	++	0
Prose drafting from author outlines	execution	0	0	+
Proof reformulation from GPT-5.4 drafts	execution	0	0	+
LaTeX typesetting of definitions	execution	0	0	+

The absence of AI ++ rows reflects the structure of the collaboration accurately. The AI never proposed a theorem, connection, or framing that was not first requested by the author. The closest candidates—the prose formulation of the traversal–UAT connection and the K&S remark—were both responses to explicit author prompts, and so receive + rather than ++. Any attribution of ++ to the AI in this paper would misrepresent the collaboration.